Bridging between in silico prediction and in vitro functional characterisation of RNA secondary structures

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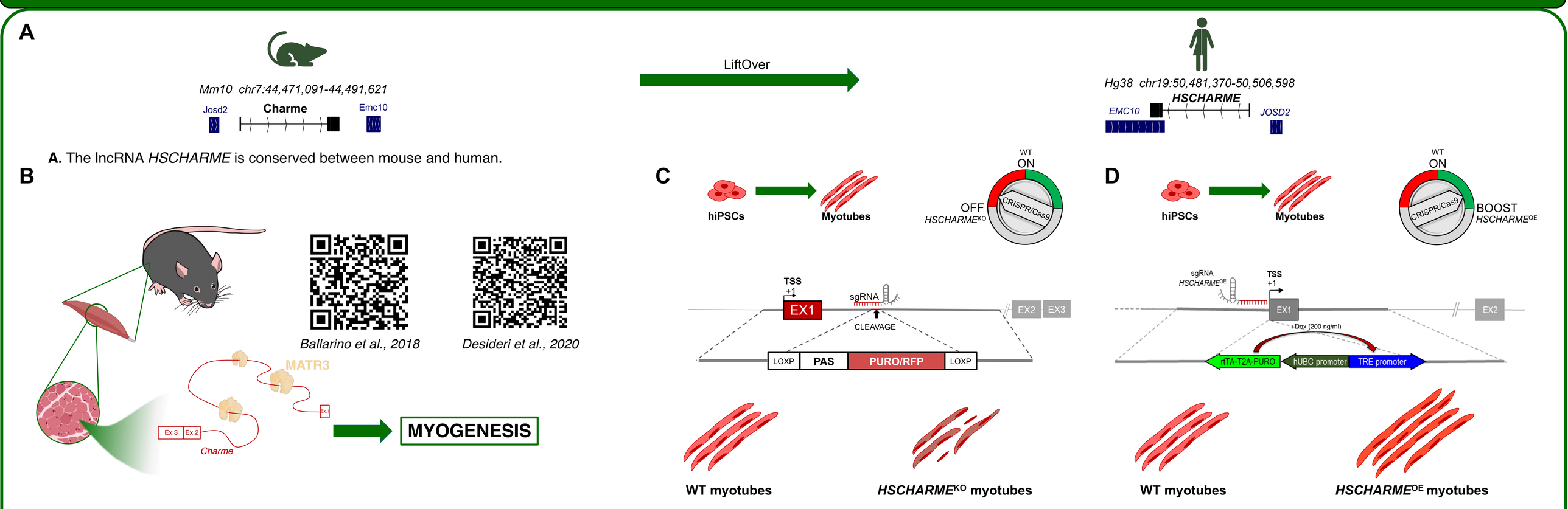
Long non-coding RNA (lncRNA) | RNA–protein interactions | RNA secondary structure prediction | CRISPR/Cas9 functional validation

Abstract

Tissue-specific long non-coding RNAs (lncRNAs) play pivotal roles in cellular physiology and differentiation, although the mechanisms of their action remain largely elusive. In particular, their ability to interact with RNA-binding proteins (RBPs) enables them to coordinate crucial cellular functions. We are currently investigating the human lncRNA *HSCHARME* (Buonaiuto et al., 2025), which appears to play a critical role during human myogenesis. In particular, we are focusing on disentangling MATR3 binding along the *HSCHARME* intron 1, which spans over 12 kb in length, and on understanding the structural determinants underlying this interaction. We predicted secondary structures along the *pCHARME* sequence by using different web-based prediction tools (such as Ipknot++, RNAFold, RNAStructure, Pkiss), using sliding windows of various sizes.
Focusing, then, on the primary sequence between nucleotide 3874 and 7914 of the intron (which contains a big proportion of the consensus motifs for MATR3 binding), we obtained a consensus secondary structure from the predictions for *pCHARME* intron 1 using a Python script. After mapping all the motifs for MATR3 binding, we predicted a 3D structure of these regions to perform docking simulations. Meanwhile, we are moving in vitro to remove, through CRISPR/Cas9 technology, pCharme intron1 and analyse how this affects its binding to MATR3 and its role in myogenesis.
Then, we plan to refine the analysis by selectively deleting the above-identified region of the intron using CRISPR/Cas9 and subsequently confirm its role in directing the *pCHARME*-MATR3 interaction in vitro.

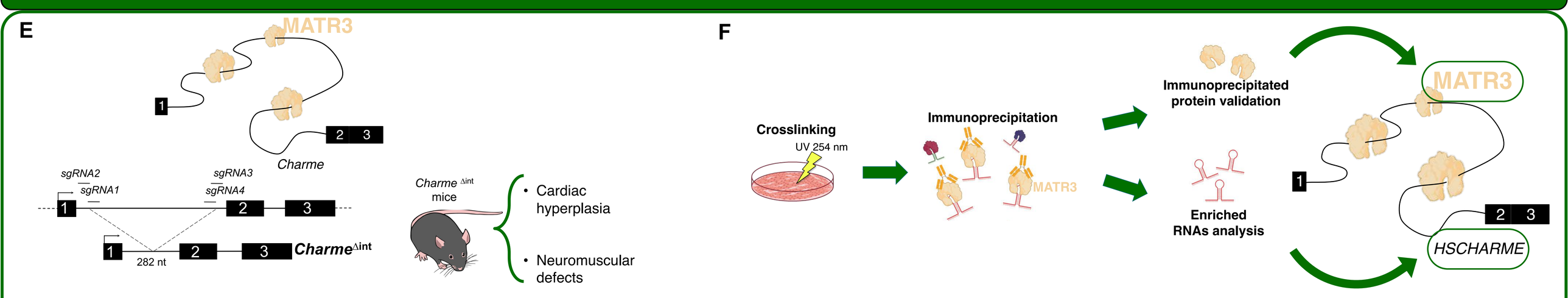
LncRNA *HSCHARME* is involved in myogenesis



B. The murine lncRNA *HSCHARME* exerts a **pro-myogenic** role which has been previously characterised.

Modulating through CRISPR/Cas9 *HSCHARME* levels in hiPSCs, **human myogenesis** is affected *in vitro*, being impaired in the **knock-out** cell line (**C**) and improved in the **overexpressing** one (**D**) (*manuscript in preparation*) .

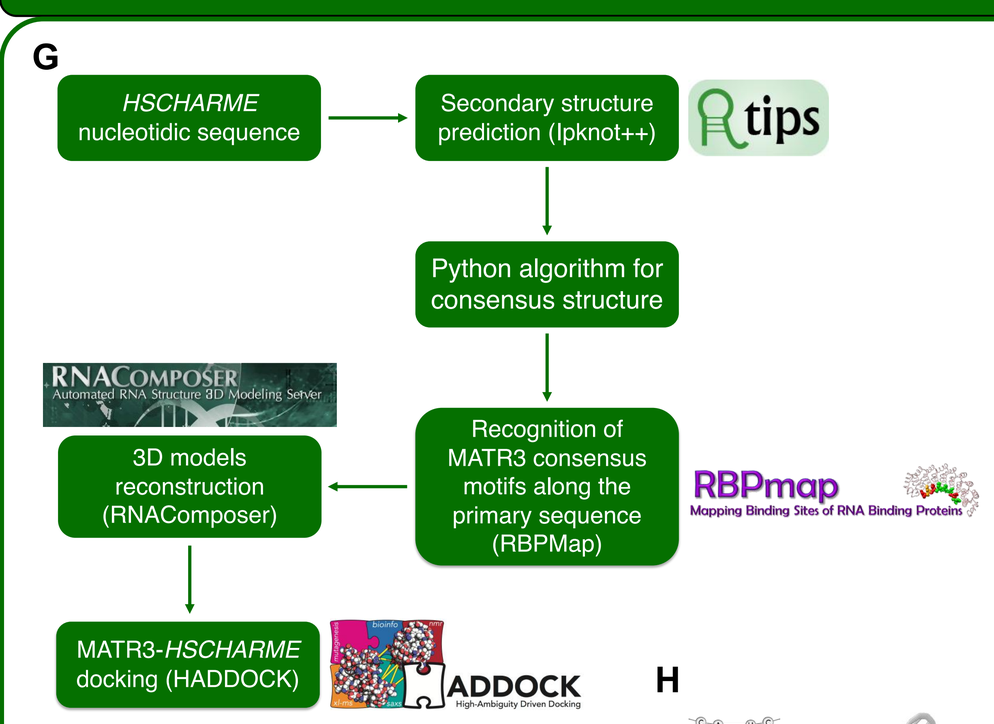
HSCHARME binds MATR3 through intron 1



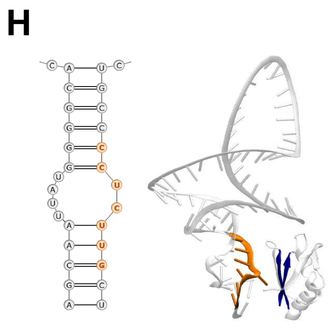
E. MATR3 binding is important for *Charme*'s role, as shown by a mouse model lacking the majority of *Charme* intron 1 (*Charme*$^{\Delta int}$) (*Desideri et al., 2020*), through which it interacts with MATR3.

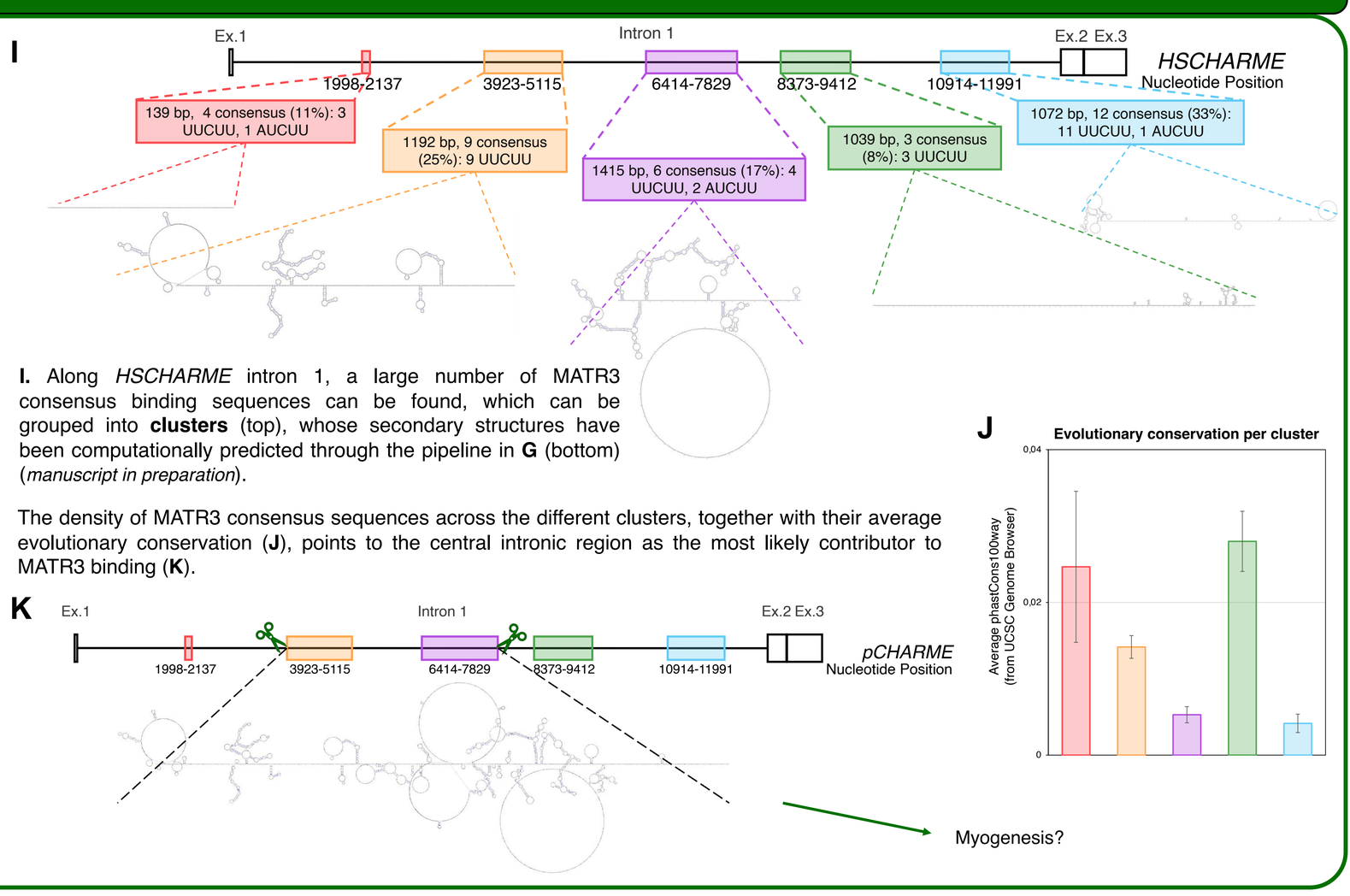
F. A **CLIP-RT-qPCR** experiment shows that the human ***HSCHARME*** conserves the ability to bind human RNA-binding protein **MATR3** (*manuscript in preparation*).

Unraveling the structural basis of *HSCHARME*–MATR3 interaction



G. Pipeline for model construction of secondary and tertiary structure of *HSCHARME*.
We produced a 2D model based on a consensus structure built upon predictions from Ipknot++ tool with and without pseudoknots.
This structure is then used to construct 3D models of the MATR3 consensus motifs along the intron 1 sequence, that are used for the docking simulations (**H**).





I. Along *HSCHARME* intron 1, a large number of MATR3 consensus binding sequences can be found, which can be grouped into **clusters** (top), whose secondary structures have been computationally predicted through the pipeline in **G** (bottom) (*manuscript in preparation*).

The density of MATR3 consensus sequences across the different clusters, together with their average evolutionary conservation (**J**), points to the central intronic region as the most likely contributor to MATR3 binding (**K**).

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